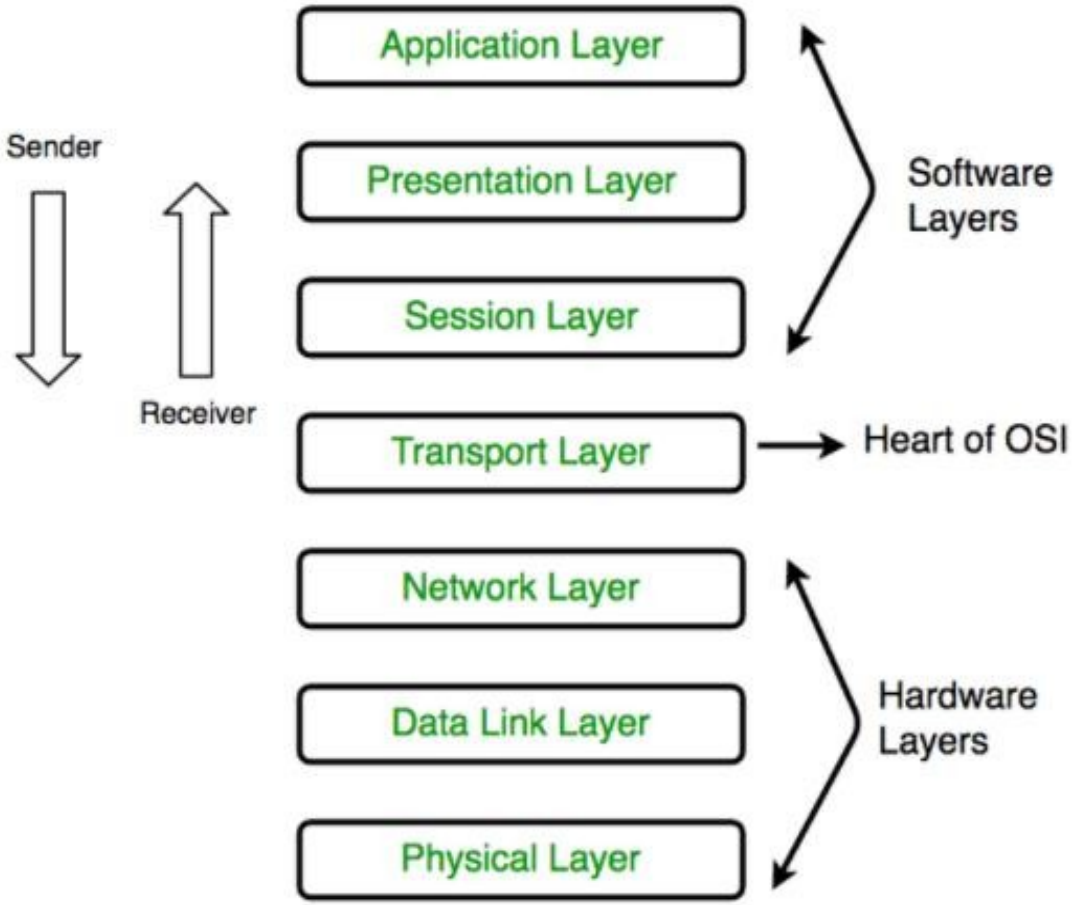
OSI Model

OSI (Open Systems Interconnection) Model



PROTOCOLS OF OSI LAYER

Physical Layer –Bluetooth, Ethernet (IEEE 802.3)

Data link Layer - PPP, HDLC

Network Layer – IP, ICMP(Internet Control Message Protocol)

Transport Layer – TCP, UDP, SCTP(Stream Control Transmission Protocol)

Session Layer –

ZIP (Zone information protocol), PAP(password authentication protocol), SCP(session control protocol)

Presentation Layer – XDR(eXternal Data Representation), TLS(encryption-authentication-integrity), SSL(client-server authentication) and MIME(Multipurpose Internet Mail Extensions)

Application Layer - HTTP, SMTP, DNS, FTP, Telnet

Physical Layer

- The lowest layer of the OSI reference model is the physical layer.
- It is responsible for the actual physical connection between the devices.
- The physical layer contains information in the form of **bits.**

Functions of Physical Layer

1. **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at bit level.

2. **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.

3. **Physical topologies:** Physical layer specifies the way in which the different, devices/nodes are arranged in a network i.e. bus, star or mesh topology.

4. **Transmission mode:** Physical layer also defines the way in which the data flows between the two connected devices. The various transmission modes possible are: Simplex, half-duplex and full-duplex.

Data Link Layer (DLL)

- The data link layer is responsible for the node to node delivery of the message.
- The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.
- When a packet arrives in a network, it is the responsibility of DLL to transmit it to the Host using its MAC address.
- Data Link Layer is divided into two sub layers :

1. Logical Link Control (LLC)
2. Media Access Control (MAC)

DLL

- The packet received from Network layer is further divided into frames depending on the frame size of NIC(Network Interface Card).
- DLL also encapsulates Sender and Receiver’s MAC address in the header.
- The Receiver’s MAC address is obtained by placing an ARP(Address Resolution Protocol) request.

Functions of DLL

1. **Framing:** Framing provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.

2. **Physical addressing:** After creating frames, Data link layer adds physical addresses (MAC address) of sender and/or receiver in the header of each frame.

3. **Error control:** Error control mechanism detects and retransmits damaged or lost frames.

4. **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus , flow control coordinates that amount of data that can be sent before receiving acknowledgement.

5. **Access control:** When a single communication channel is shared by multiple devices, MAC sub-layer of data link layer helps to determine which device has control over the channel at a given time.

Network Layer

- Network layer works for the transmission of data from one host to the other located in different networks.
- It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available.
- The sender & receiver's IP address are placed in the header by the network layer.

Functions of Network Layer

1. **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of network layer is known as routing.

2. **Logical Addressing:** In order to identify each device on internetwork uniquely, network layer defines an addressing scheme. The sender & receiver's IP address are placed in the header by network layer.

Transport Layer

- Transport layer provides services to application layer and takesservices from network layer.
- It is responsible for the End to End Delivery of the complete message.
- The transport layer also provides the acknowledgement of the successful data transmission and re-transmits the data if an error is found.
- *Data in the Transport Layer is called as* ***Segments****.*

Transport Layer at Sender Side

- Transport layer receives the formatted data from the upper layers, performs **Segmentation** and also implements **Flow & Error control** to ensure proper data transmission.
- It also adds Source and Destination port number in its header and forwards the segmented data to the Network Layer.
- The sender need to know the port number associated with the receiver's application. (well known / ephemeral port No.)

Transport Layer at Receiver Side

- Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application.
- It also performs sequencing and reassembling of the segmented data.

Functions of Transport Layer

1. **Segmentation and Reassembly:** This layer accepts the message from the (session) layer , breaks the message into smaller units . Each of the segment produced has a header associated with it. The transport layer at the destination station reassembles the message.
2. **Service Point Addressing:** In order to deliver the message to correct process, transport layer header includes a type of address called service point address or **port** address. Thus by specifying this address, transport layer makes sure that the message is delivered to the correct process.

Services provided by TL

1. **Connection Oriented Service:** It is a three-phase process which include
   – Connection Establishment
   – Data Transfer
   – Termination / disconnection

   In this type of transmission, the receiving device sends an acknowledgement, back to the source after a packet or group of packet is received. This type of transmission is reliable and secure. **(TCP)**
2. **Connection less service:** It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service. **(UDP)**

Session Layer

- This layer is responsible for establishment of connection, maintenance of sessions, authentication and also ensures security.

- The functions of the session layer are :

1. **Session establishment, maintenance and termination:** The layer allows the two processes to establish, use and terminate a connection.
2. **Synchronization :** This layer allows a process to add checkpoints which are considered as synchronization points into the data. These synchronization point help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
3. **Dialog Controller :** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Presentation Layer

- Presentation layer is also called the **Translation layer**.
- The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

- The functions of the presentation layer are :

1. **Translation :** For example, ASCII to EBCDIC.
2. **Encryption/ Decryption :** Data encryption translates the data into another form or code. The encrypted data is known as the cipher text and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
3. **Compression:** Reduces the number of bits that need to be transmitted on the network.

Application Layer

- At the very top of the OSI Reference Model stack of layers, we find Application layer which is implemented by the network applications.
- These applications produce the data, which has to be transferred over the network.
- This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

Ex: Application – Browsers, Skype Messenger etc.

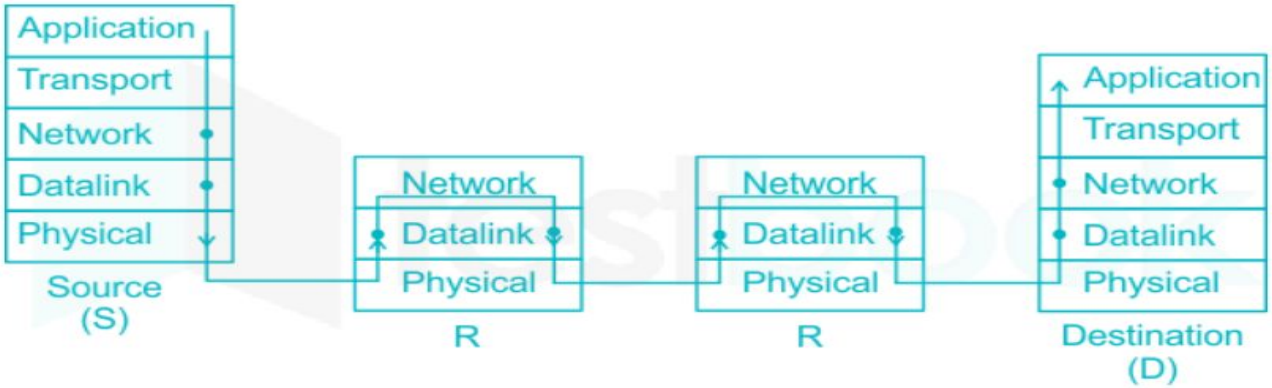
Functions of Application Layer

1. Network Virtual Terminal
2. FTAM-File transfer access and management
3. Mail Services
4. Directory Services

Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.

Answer

Explanation:



At Intermediate Router R the packet will not go beyond the network layer. Hence Network layer will only be visited once at both the routers.

But the Data link layer will be visited twice.

Hence Network layer will be visited 4 times and the Data link layer will be visited 6 times (1 + 2 (at R) + 2 (at R) + 1 = 6)

Connecting devices available in layers

- Hub : Physical layer
- Switch: Physical & data link layer
- Routers: Physical, data link & network layer
- Bridge: Data link layer
- Repeaters : Physical layer